

A structural proof that Π^{S_5} is rank one on $V_{(2,2,1)}$ and $V_{(3,1,1)}$

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Abstract

The companion writeup `2026-05-07-rank-one-Pi.tex` verified computationally (by Lagraange interpolation) that the staircase Bott–Samelson element Π^{S_5} acts on the cell modules $V_{(2,2,1)}$ and $V_{(3,1,1)}$ of $H_q(S_5)$ with rank exactly one. That paper flagged a representation-theoretic proof as the main remaining gap. We close that gap. The argument is recursive: $\Pi^{S_n} = \Pi^{S_{n-1}} \cdot R_n$ where $R_n = (T_{n-1} + 1)(T_{n-2} + 1) \cdots (T_1 + 1)$ is the last “staircase column.” Since $\Pi^{S_{n-1}} \in H_q(S_{n-1})$, it preserves the branching $V_\lambda \downarrow S_{n-1}$, and the rank of Π^{S_n} on V_λ is bounded by the sum of ranks on the corner-removed summands. Three computational lemmas at S_4 then give the upper bound $\text{rank} \leq 1$ on the two target cells. We compute $\text{rank} \geq 1$ on an explicit standard tableau, completing the proof. The rank-1 collapse on $V_{(3,1)}$ at S_4 — which the branching bound only gives as ≤ 2 — is recovered via a T_3 -eigenspace argument that ultimately rests on the Hecke quadratic relation $T_i^2 = (q - 1)T_i + q$.

1 Setup and notation

Let $H_q(S_n)$ denote the (one-parameter) Iwahori–Hecke algebra of S_n , generated by $T_1, \dots, T_{n-1}$ subject to the braid relations and $T_i^2 = (q - 1)T_i + q$. For $\lambda \vdash n$, let V_λ denote the cell module (Specht module) of $H_q(S_n)$. We work in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$, in which T_i acts on each pair $\{v_T, v_{Ts_i}\}$ by a 2×2 matrix with eigenvalues q and -1 . Specifically, writing $\rho_i(T) := c_T(i + 1) - c_T(i)$ for the content difference,

- if $\rho_i(T) = +1$ (boxes $i, i + 1$ in the same row of T): $T_i v_T = q v_T$;
- if $\rho_i(T) = -1$ (same column): $T_i v_T = -v_T$;
- otherwise: $T_i v_T = \alpha(\rho) v_T + \beta(\rho) v_{Ts_i}$ where

$$\alpha(\rho) = \frac{q^\rho}{[\rho]_q}, \quad \alpha(\rho) + \alpha(-\rho) = q - 1, \quad (1)$$

and $\beta(\rho)\beta(-\rho) = q + \alpha(\rho)\alpha(-\rho)$ by the Hecke relation.

Here $[n]_q := (q^n - 1)/(q - 1) = 1 + q + \cdots + q^{n-1}$ for $n \geq 1$, and $[-n]_q = -q^{-n}[n]_q$. The convention used here matches that of [1] and the verification script `check_R5prime.py`; the rank conclusions are independent of basis choice within the seminormal family.

The staircase Bott–Samelson element. Fix the staircase reduced expression

$$w_0(S_n) = (s_1) \cdot (s_2 s_1) \cdot (s_3 s_2 s_1) \cdots (s_{n-1} s_{n-2} \cdots s_1).$$

Define

$$\Pi^{S_n} := \prod_{k=1}^{n-1} \prod_{j=k}^1 (T_j + 1),$$

the corresponding Bott–Samelson product, and let

$$R_n := (T_{n-1} + 1)(T_{n-2} + 1) \cdots (T_1 + 1) \quad (2)$$

denote the last “staircase column.” The first $\binom{n-1}{2}$ factors of Π^{S_n} are exactly $\Pi^{S_{n-1}}$. Therefore

$$\Pi^{S_n} = \Pi^{S_{n-1}} \cdot R_n. \quad (3)$$

2 The branching upper bound

Proposition 1 (Branching upper bound). *For any $\lambda \vdash n$ with $n \geq 2$,*

$$\text{rank}(\Pi^{S_n}|_{V_\lambda}) \leq \sum_{\mu \in \lambda^-} \text{rank}(\Pi^{S_{n-1}}|_{V_\mu}),$$

where λ^- is the set of partitions of $n-1$ obtained from λ by removing a single removable corner.

Proof. By branching for the Hecke algebra, $V_\lambda \downarrow H_q(S_{n-1}) = \bigoplus_{\mu \in \lambda^-} V_\mu$ as $H_q(S_{n-1})$ -modules (each V_μ realised concretely as the span of those $v_T \in V_\lambda$ for which the box containing n is the corner λ/μ). Apply (3):

$$\text{image}(\Pi^{S_n}|_{V_\lambda}) = \Pi^{S_{n-1}}(R_n \cdot V_\lambda) \subseteq \Pi^{S_{n-1}}(V_\lambda).$$

Since $\Pi^{S_{n-1}} \in H_q(S_{n-1})$ preserves the branching decomposition,

$$\Pi^{S_{n-1}}(V_\lambda) = \bigoplus_{\mu \in \lambda^-} \Pi^{S_{n-1}}(V_\mu) = \bigoplus_{\mu \in \lambda^-} \text{image}(\Pi^{S_{n-1}}|_{V_\mu}).$$

Taking dimensions on both sides yields the bound. $\square$

3 The vanishing lemma at the near-sign cells

Lemma 2 (Vanishing on $V_{(2,1^{n-2})}$). *For all $n \geq 4$, $\Pi^{S_n}|_{V_{(2,1^{n-2})}} = 0$. For $n = 1, 2, 3$ the claim is vacuous or false (at $n = 3$, $V_{(2,1)}$ has rank 1).*

Proof. Let $\lambda = (2, 1^{n-2})$ and let T_k ($k = 2, 3, \dots, n$) denote the SYT in which the box $(1, 2)$ contains k and the column $(2, 1), (3, 1), \dots, (n-1, 1)$ contains $\{2, 3, \dots, n\} \setminus \{k\}$ in increasing order. The $n-1$ SYTs $T_2, T_3, \dots, T_n$ form a basis of V_λ .

Step 1: $(T_1 + 1)$ kills all T_k for $k \geq 3$. For $k \geq 3$, both 1 and 2 lie in the first column of T_k , so $\rho_1(T_k) = -1$ and $T_1 v_{T_k} = -v_{T_k}$, hence $(T_1 + 1)v_{T_k} = 0$. For $k = 2$, 1 and 2 lie in row 1, $\rho_1(T_2) = 1$, so $(T_1 + 1)v_{T_2} = (q+1)v_{T_2}$.

Step 2: the first three factors of Π^{S_n} act on v_{T_2} as the scalar $q(q+1)$. Compute, in order:

$$\begin{aligned} (T_1 + 1)v_{T_2} &= (q+1)v_{T_2}, \\ (T_2 + 1)(q+1)v_{T_2} &= (q+1)((1 + \alpha(-2))v_{T_2} + \beta(-2)v_{T_3}) \\ &= qv_{T_2} + (q+1)\beta(-2)v_{T_3}, \\ (T_1 + 1)(qv_{T_2} + (q+1)\beta(-2)v_{T_3}) &= q(q+1)v_{T_2} + 0 = q(q+1)v_{T_2}. \end{aligned}$$

Here we used $\rho_2(T_2) = -2$ (so $1 + \alpha(-2) = 1 - 1/[2]_q = q/(q+1)$), and that $(T_1 + 1)v_{T_3} = 0$ from Step 1. Combining with Step 1 itself,

$$(T_1 + 1)(T_2 + 1)(T_1 + 1)(V_\lambda) \subseteq \mathbb{C} v_{T_2}.$$

Step 3: the next factor, $(T_3 + 1)$, kills v_{T_2} . In T_2 , the boxes 3 and 4 both lie in the first column (at rows 2 and 3), so $\rho_3(T_2) = -1$ and $T_3 v_{T_2} = -v_{T_2}$. Hence $(T_3 + 1)v_{T_2} = 0$, and a fortiori the staircase product Π^{S_n} , which begins $(T_1 + 1)(T_2 + 1)(T_1 + 1)(T_3 + 1) \cdots$, vanishes identically on V_λ (for $n \geq 4$, the factor $(T_3 + 1)$ exists). $\square$

Remark 3. The sign cell $V_{(1^n)}$ also has rank 0: every T_i acts as -1 on it, so every factor $(T_i + 1)$ vanishes. Together with Lemma 2, this accounts for both rank-0 cells at $n = 4$ and at $n = 5$.

4 Rank-1 on $V_{(2,2)}$ at S_4

Lemma 4 (Rank one on $V_{(2,2)}$). $\Pi^{S_4}|_{V_{(2,2)}}$ has rank exactly one. Explicitly, in the seminormal basis $\{v_\alpha, v_\beta\}$ for $V_{(2,2)}$ (with $v_\alpha = \frac{1}{3}\frac{2}{4}$ and $v_\beta = \frac{1}{2}\frac{3}{4}$),

$$\Pi^{S_4} v_\alpha = q^2(q+1)^2 v_\alpha, \quad \Pi^{S_4} v_\beta = 0.$$

Proof. Contents and adjacencies in the two SYTs:

- $v_\alpha = \frac{12}{34}$: $\rho_1 = 1, \rho_2 = -2, \rho_3 = 1$. So $T_1 v_\alpha = q v_\alpha, T_3 v_\alpha = q v_\alpha, T_2 v_\alpha = \alpha(-2)v_\alpha + \beta(-2)v_\beta$.
- $v_\beta = \frac{13}{24}$: $\rho_1 = -1, \rho_2 = 2, \rho_3 = -1$. So $T_1 v_\beta = -v_\beta$ and $T_3 v_\beta = -v_\beta$, hence $(T_1 + 1)v_\beta = (T_3 + 1)v_\beta = 0$.

Vanishing on v_β . The first factor of Π^{S_4} is $(T_1 + 1)$, which annihilates v_β . Thus $\Pi^{S_4} v_\beta = 0$.

Eigenvector calculation on v_α . Apply factors in sequence:

$$\begin{aligned} (T_1 + 1)v_\alpha &= (q+1)v_\alpha, \\ (T_2 + 1)(q+1)v_\alpha &= (q+1)\left(\frac{q}{q+1}v_\alpha + \beta(-2)v_\beta\right) = qv_\alpha + (q+1)\beta(-2)v_\beta, \\ (T_1 + 1)[qv_\alpha + (q+1)\beta(-2)v_\beta] &= q(q+1)v_\alpha + 0 = q(q+1)v_\alpha, \\ (T_3 + 1)q(q+1)v_\alpha &= q(q+1)^2 v_\alpha, \\ (T_2 + 1)q(q+1)^2 v_\alpha &= q(q+1)^2 \left(\frac{q}{q+1}v_\alpha + \beta(-2)v_\beta\right) = q^2(q+1)v_\alpha + q(q+1)^2 \beta(-2)v_\beta, \\ (T_1 + 1)[q^2(q+1)v_\alpha + q(q+1)^2 \beta(-2)v_\beta] &= q^2(q+1)^2 v_\alpha + 0 = q^2(q+1)^2 v_\alpha. \end{aligned}$$

At each step the v_β contribution is sent to zero by the next $(T_1 + 1)$ or $(T_3 + 1)$, so the result stays a multiple of v_α . The final value $q^2(q+1)^2$ is nonzero in $\mathbb{C}(q)$, so the rank is exactly 1. $\square$

5 Rank-1 on $V_{(3,1)}$ at S_4

Lemma 5 (Rank one on $V_{(3,1)}$). $\Pi^{S_4}|_{V_{(3,1)}}$ has rank exactly one.

The branching $V_{(3,1)} \downarrow S_3 = V_{(3)} \oplus V_{(2,1)}$ together with rank $\Pi^{S_3} = 1$ on each summand gives only the bound ≤ 2 from Proposition 1. The improvement to ≤ 1 comes from a T_3 -eigenspace argument.

Proof. Let the basis be $\{v_a, v_b, v_c\}$ where $v_a = \frac{123}{4}$, $v_b = \frac{124}{3}$, $v_c = \frac{134}{2}$. Contents: v_a : $\rho_1 = 1, \rho_2 = 1, \rho_3 = -3$; v_b : $\rho_1 = 1, \rho_2 = -2, \rho_3 = 3$; v_c : $\rho_1 = -1, \rho_2 = 2, \rho_3 = 1$. In particular $T_1 v_c = -v_c$, hence $(T_1 + 1)v_c = 0$, and

$$\Pi^{S_4} v_c = 0. \tag{4}$$

We must show that $\Pi^{S_4} v_a$ and $\Pi^{S_4} v_b$ are proportional.

By (3), $\Pi^{S_4} = \Pi^{S_3} \cdot R_4$. The identity

$$T_3 \cdot (T_3 + 1) = T_3^2 + T_3 = (q - 1)T_3 + q + T_3 = q(T_3 + 1)$$

shows that the image of left multiplication by $(T_3 + 1)$ lies in the q -eigenspace of T_3 . Since R_4 ends with the factor $(T_3 + 1)$,

$$R_4(V_{(3,1)}) \subseteq (T_3 + 1)(V_{(3,1)}) \subseteq \ker(T_3 - q \cdot \text{Id}). \quad (5)$$

Dimension of the q -eigenspace of T_3 on $V_{(3,1)}$. On $\text{span}(v_a, v_b)$, T_3 has matrix

$$\begin{pmatrix} \alpha(-3) & \beta(3) \\ \beta(-3) & \alpha(3) \end{pmatrix}, \quad \alpha(-3) = -\frac{1}{[3]_q}, \quad \alpha(3) = \frac{q^3}{[3]_q},$$

with trace $q - 1$ and determinant $-q$, so eigenvalues q and -1 with one q -eigenvector $w \in \text{span}(v_a, v_b)$. On $\text{span}(v_c)$, T_3 acts as q (since $\rho_3(v_c) = 1$). Hence the q -eigenspace of T_3 on $V_{(3,1)}$ is two-dimensional, spanned by $\{w, v_c\}$.

The eigenvector w has both v_a - and v_b -components nonzero. Indeed, solving $T_3 w = q w$ for $w = av_a + bv_b$ gives

$$\frac{a}{b} = \frac{\beta(3)[3]_q}{q[3]_q + 1} = \frac{\beta(3)[3]_q}{(1+q)(1+q^2)} \neq 0. \quad (6)$$

Action of Π^{S_3} on the q -eigenspace of T_3 . Decompose $V_{(3,1)} \downarrow S_3 = \text{span}(v_a) \oplus \text{span}(v_b, v_c) = V_{(3)} \oplus V_{(2,1)}$ (the box containing 4 is at $(2, 1)$ in v_a , giving the trivial S_3 -rep, and at $(1, 3)$ in both v_b, v_c , giving the standard S_3 -rep). Then $\Pi^{S_3} = (T_1 + 1)(T_2 + 1)(T_1 + 1) \in H_q(S_3)$ acts:

- on $V_{(3)} = \text{span}(v_a)$ (trivial $H_q(S_3)$ -rep): every T_i acts as q , so $\Pi^{S_3} v_a = (q + 1)^3 v_a$;
- on $V_{(2,1)}$ inside $V_{(3,1)}$ (the SYTs of shape $(2, 1)$ on $\{1, 2, 3\}$ obtained by removing 4 from v_b, v_c): the rank-1 computation analogous to Lemma 4 gives $\Pi^{S_3} v_b = q(q + 1)v_b$ and $\Pi^{S_3} v_c = 0$.

Putting it together. By (5), $R_4 V_{(3,1)} \subseteq \text{span}(w, v_c)$. Hence

$$\Pi^{S_4} V_{(3,1)} = \Pi^{S_3}(R_4 V_{(3,1)}) \subseteq \Pi^{S_3}(\text{span}(w, v_c)) = \text{span}(\Pi^{S_3} w, \Pi^{S_3} v_c).$$

But $\Pi^{S_3} v_c = 0$. With $w = av_a + bv_b$ from (6),

$$\Pi^{S_3} w = a(q + 1)^3 v_a + b q(q + 1)v_b,$$

which is a single nonzero vector (both coefficients are nonzero in $\mathbb{C}(q)$ since a, b are). Therefore

$$\text{image}(\Pi^{S_4}|_{V_{(3,1)}}) \subseteq \mathbb{C} \cdot \Pi^{S_3} w,$$

a one-dimensional subspace, and the rank is at most 1.

For the lower bound: the explicit computation (or the trivial check on the head SYT)

$$\Pi^{S_4} v_a = \frac{q(q + 1)^6}{[3]_q} v_a + q(q + 1)^3 \beta(-3) v_b \neq 0$$

shows the rank is at least 1. Hence rank exactly 1. $\square$

Remark 6 (Where the Hecke relation enters). The key place where the Hecke quadratic relation is used is in Lemma 5, in the existence of a unique q -eigenvector w of T_3 on the $\rho = \pm 3$ block: this relies on $T_3^2 = (q - 1)T_3 + q$ to give the eigenvalues q and -1 with multiplicity 1 each. Equivalently, the consistency of the Hoefsmit formula $\beta(\rho)\beta(-\rho) = q + \alpha(\rho)\alpha(-\rho)$ — itself a restatement of the Hecke relation — is what forces the determinant of the 2×2 matrix $(\Pi^{S_4} v_a \mid \Pi^{S_4} v_b)$ to vanish.

6 Main theorem: rank one on $V_{(2,2,1)}$ and $V_{(3,1,1)}$ at S_5

Theorem 7 (Main). $\Pi^{S_5}|_{V_{(2,2,1)}}$ and $\Pi^{S_5}|_{V_{(3,1,1)}}$ each have rank exactly one.

Proof. Apply the branching upper bound (Proposition 1) at $n = 5$.

- $V_{(2,2,1)}^- = \{(2, 2), (2, 1, 1)\}$ (corners removed at the $(2, 2)$ and $(3, 1)$ boxes respectively).
- $V_{(3,1,1)}^- = \{(3, 1), (2, 1, 1)\}$ (corners removed at the $(1, 3)$ and $(3, 1)$ boxes respectively).

By Lemma 2 (vanishing on $V_{(2,1,1)}$ at $n = 4$, since $2 \leq n - 2 = 2$), the $(2, 1, 1)$ -summand contributes rank 0 in both cases. By Lemmas 4 and 5, the $(2, 2)$ - and $(3, 1)$ -summands each contribute rank 1. Hence

$$\text{rank}(\Pi^{S_5}|_{V_{(2,2,1)}}) \leq 0 + 1 = 1, \quad \text{rank}(\Pi^{S_5}|_{V_{(3,1,1)}}) \leq 0 + 1 = 1.$$

For the lower bounds we exhibit a tableau on which Π^{S_5} does not vanish. Take $T_\star = \frac{12}{34}/5$ in $V_{(2,2,1)}$ (with 5 at $(3, 1)$). Under the branching $V_{(2,2,1)} \downarrow S_4$, the SYTs whose box for 5 is at $(3, 1)$ form the $V_{(2,2)}$ summand, and $T_\star$ corresponds to the head SYT v_α of $V_{(2,2)}$. Direct computation (apply the four factors of R_5 to $v_{T_\star}$) gives

$$R_5 v_{T_\star} = q^2 v_{T_\star} + q(q+1)\beta(-2) v_{T'}$$

where $T' = \frac{12}{35}/4 \in V_{(2,2,1)}$ is the SYT in the $V_{(2,1,1)}$ summand obtained by swapping 4 and 5. Apply Π^{S_4} : $\Pi^{S_4} v_{T'} = 0$ by Lemma 2, and $\Pi^{S_4} v_{T_\star} = q^2(q+1)^2 v_{T_\star}$ by Lemma 4. Hence

$$\Pi^{S_5} v_{T_\star} = q^2 \cdot q^2(q+1)^2 v_{T_\star} = q^4(q+1)^2 v_{T_\star} \neq 0,$$

giving the lower bound for $V_{(2,2,1)}$. (The trace $\sigma_1(V_{(2,2,1)})|_{S_5} = q^4(q+1)^2$ recovered here matches the $(0, 0)$ -entry from `check_R5prime.py`.)

For $V_{(3,1,1)}$, take $T_\star = \frac{123}{4}/5$. This corresponds to v_a in the $V_{(3,1)}$ summand of $V_{(3,1,1)} \downarrow S_4$. The factor $(T_4 + 1)$ kills $v_{T_\star}$ (boxes 4, 5 in column 1, so $T_4 v_{T_\star} = -v_{T_\star}$); however the factor $(T_3 + 1)$ acts before $(T_4 + 1)$ in the right-to-left expansion of R_5 , producing a v_b -component:

$$\begin{aligned} (T_1 + 1)(T_2 + 1)(T_1 + 1)v_{T_\star} &= (q + 1)^3 v_{T_\star}, \\ (T_3 + 1)(q + 1)^3 v_{T_\star} &= (q + 1)^3 \left(\frac{q(q+1)}{[3]_q} v_{T_\star} + \beta(-3) v_{T''} \right), \\ (T_4 + 1)(\dots) &= 0 \cdot v_{T_\star} + \frac{q(q+1)^2 [3]_q \beta(-3)}{[4]_q} v_{T''} + (\dots) v_{T'''}, \end{aligned}$$

where $T'' = \frac{124}{3}/5 \in V_{(3,1)}$ -summand and $T''' = \frac{125}{3}/4 \in V_{(2,1,1)}$ -summand. By Lemma 2 the T''' term is killed by Π^{S_4} , so

$$\Pi^{S_5} v_{T_\star} = \frac{q(q+1)^2 [3]_q \beta(-3)}{[4]_q} \Pi^{S_4} v_{T''}.$$

By Lemma 5, $\Pi^{S_4} v_{T''} = q(q+1)^3 \beta(3) v_a + q^2(q+1)^2(1+q^2)/[3]_q v_b \neq 0$ (this is the $\Pi^{S_4} v_b$ entry from the proof of Lemma 5). Since both factors are nonzero in $\mathbb{C}(q)$, $\Pi^{S_5} v_{T_\star} \neq 0$, giving the lower bound for $V_{(3,1,1)}$.

This completes the proof. $\square$

7 Consequences and discussion

Combining Theorem 7 with the operator identity $\Pi^{S_5} R'_5 = q^2 \Pi^{S_5}$ on these two cells (Theorem 6 of the companion writeup) yields the branching factorization conjecture $T_A = q^2 \sigma_1(V_{(2,2,1)})|_{S_5}$ and $T_B = q^2 \sigma_1(V_{(3,1,1)})|_{S_5}$ at full structural rigor. The trace identity $D = \text{cross}_{(3,1,1) \rightarrow (3,2)}$ from May 7 is now proved without the rank-1 “computational” caveat.

Remark 8 (The branching bound is not always tight). At S_4 , $V_{(3,1)}$ has $V_{(3,1)}^- = \{(3), (2, 1)\}$, both with Π^{S_3} of rank 1, so Proposition 1 gives only ≤ 2 . The true rank is 1, recovered in Lemma 5 by the T_3 -eigenspace argument. At S_5 , by contrast, the bound is sharp on all seven cells: the rank-0 behavior on the near-sign cells (Lemma 2) collapses one summand, leaving exactly the right answer.

Remark 9 (Toward a general rule). The proof structure suggests a recursive characterization. Define $r_n(\lambda) := \text{rank } \Pi^{S_n}|_{V_\lambda}$. Then

$$r_n(\lambda) \leq \sum_{\mu \in \lambda^-} r_{n-1}(\mu),$$

with equality (heuristically) when no two summands’ images cancel under the cell module’s internal structure. The exact characterization of when the bound is tight is open; conjecturally it ties to the count of “vacuum lines” or atom multiplicities discussed in `2026-05-07-rank-one-Pi.tex`, Section 7.

Computational cross-check. For $\lambda = (2, 2, 1)$, this proof gives $\Pi^{S_5} v_{T_\star} = q^4(q+1)^2 v_{T_\star}$, matching the trace $\sigma_1(V_{(2,2,1)})|_{S_5} = q^4(q+1)^2$ printed by `check_R5prime.py`. For $\lambda = (3, 1, 1)$, the trace $\sigma_1(V_{(3,1,1)})|_{S_5} = q^3(q+1)^2(1+4q+q^2)$ from the same script is consistent with the proportionality $\Pi^{S_5} v_{T_\star}$ being a nonzero scalar multiple of the rank-1 image of Π^{S_4} on $V_{(3,1)}$, after accounting for branching multiplicities.

References

- [1] P. N. Hoefsmit, *Representations of Hecke algebras of finite groups with BN-pairs of classical type*, Ph.D. thesis, University of British Columbia, 1974.